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Roll No

TX-5002-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Measurement and Control

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the uncertainty analysis in brief.
b) Discuss the various types of errors in measurement.
2. a) What is Measurement? Discuss block diagram of general measurement system.
b) Discuss Data representing format.
3. a) Discuss about GPIB electrical interface.
b) Explain the working principle of LED.
4. a) Write short note on RS232C and IEEE488.
b) Discuss advantages of digital instrument over analog instrument.
5. a) Differentiate the open loop and closed loop control system.
b) Explain in brief the transfer functions of elements, system and processes.

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6.
 - a) Describe noise with its types.
 - b) Discuss about data acquisition and data reduction with suitable diagram.
7.
 - a) What is Error? Discuss its types. How can they be reduced?
 - b) Explain higher order uncertainty analysis.
8. Write short notes on:
 - a) Calibration
 - b) Error propagation
 - c) Modeling of thermal system
